

Beyond Mean-Variance

Goal-Based Portfolio Optimisation with Derivatives and Structured Products

From a mental-accounting thesis to a scalable Monte-Carlo + CVaR engine

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Abstract. This note documents a research optimiser that builds goal-based portfolios beyond the classical mean-variance frame. It re-implements and extends a 2012 MSc thesis (USI Lugano) that replicated the mental-accounting models of Das & Statman (2009) and Das, Markowitz, Scheid & Statman (2010). Two engines are provided: an **exact grid optimiser** that reproduces the thesis results on a discretised state space, and a **scalable Monte-Carlo + CVaR optimiser** that escapes the curse of dimensionality through copula scenarios and a linear program. Both price derivatives and structured products arbitrage-consistently, and an out-of-sample back-test holds the chosen portfolio on later, unseen data. The work is a research and educational project; it is not investment advice.

1. Background and motivation

Markowitz mean-variance (MV) analysis treats risk as a single portfolio variance and assumes the investor optimises one aggregate trade-off. Behavioural evidence instead suggests that people manage wealth in **mental accounts** tied to distinct goals, each with its own attitude to loss. Das, Markowitz, Scheid and Statman (2010) formalise this and show a precise equivalence between the mental-accounting (MAT) specification and classical MV; Das and Statman (2009) extend the goal-based optimisation to **derivatives and non-normal returns**.

The author's 2012 MSc Finance thesis (USI Lugano, supervised by Prof. Enrico De Giorgi) replicated both methodologies in R and extended them with additional derivative simulations and parameter analysis. The present work re-implements that foundation in Python and extends it with live market data, a wider library of structured products, an out-of-sample back-test, and a scalable scenario-based engine.

2. The mental-accounting framework

An investor allocates wealth across goal sub-portfolios. Each goal carries a return threshold H and a maximum acceptable probability of falling short of it, α . In Value-at-Risk form the goal problem is to maximise expected return subject to a shortfall-probability constraint:

$$\max_w E[r_p] \quad \text{s.t.} \quad \Pr(r_p < H) \leq \alpha, \quad \sum_i w_i = 1$$

A tail-average (Expected-Shortfall) variant replaces the probability constraint with a floor on the conditional mean of the loss tail, $E[r_p \mid r_p < H] \geq L$. The key theoretical result is the **MVT-MAT equivalence**: each goal's optimum is mean-variance efficient and can be written with an implied risk-aversion coefficient λ ,

$$w^*(\lambda) = \arg \max_w \left\{ E[r_p] - \frac{\lambda}{2} \text{Var}(r_p) \right\}$$

so the behavioural specification can be read back as a classical MV problem with a goal-specific risk aversion. This lets the optimiser report an implied λ for every goal and reconcile the two views of the same portfolio.

3. The exact grid optimiser

Asset returns are modelled as multivariate normal with mean vector μ and covariance Σ . The engine enumerates candidate portfolios on a discretised grid: each asset's return is sampled over $\mu \pm k\sigma$ in m steps, and weights are searched on a simplex grid of resolution m' . For n assets the state space grows as roughly m^n — the result is exact and thesis-faithful, but practical only to about four assets.

Two Expected-Shortfall modes are offered. The **thesis-faithful** mode reproduces the original R behaviour, in which the refinement step enforces the VaR penalty and the ES condition is applied at grid-eligibility; the **rigorous-ES** mode instead maximises return strictly subject to $ES \geq L$ and recovers expected return the thesis method leaves on the table. Four resolutions are available — Fast, Standard and High, plus a custom **Turbo** coarse-to-fine solver that matches High-precision VaR to within a few basis points at about sixty times the speed. For five or more assets a differential-evolution search replaces the exhaustive grid.

4. Derivatives and structured products

Sixteen instruments are priced arbitrage-consistently with Black–Scholes — vanilla options, spreads, collars, capital-guaranteed and barrier notes, and certificates — alongside a builder for configuring bespoke payoffs. The Black–Scholes call and its standard inputs are

$$C = S_0 \Phi(d_1) - Ke^{-rT} \Phi(d_2), \quad d_{1,2} = \frac{\ln(S_0/K) + (r \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

Under each modelled outcome the instrument's payoff is evaluated and folded into the portfolio return, so asymmetric, non-normal payoffs enter the optimisation directly rather than through a variance proxy — precisely the behaviour the classical mean-variance frame cannot represent.

5. The scalable Monte-Carlo + CVaR engine

To escape the m^n state space, the second engine samples S joint return scenarios through a copula — Gaussian, or Student- t for tail co-movement (assets crashing together) — prices any derivatives under each scenario, and solves the goal as a **linear program**. Tail risk is measured coherently with Conditional Value-at-Risk (Expected Shortfall). The Rockafellar–Uryasev representation is

$$\text{CVaR}_\alpha = \min_{\zeta} \left\{ \zeta + \frac{1}{\alpha} E[(-r_p - \zeta)^+] \right\}$$

which is linear in scenario form. Maximising expected return subject to a CVaR floor L then becomes

$$\max_{w, \zeta, z} E[r_p] \quad \text{s.t.} \quad \zeta + \frac{1}{\alpha S} \sum_{s=1}^S z_s \leq -L, \quad z_s \geq -w^T R_s - \zeta, \quad z_s \geq 0, \quad \sum_i w_i = 1, \quad 0 \leq w \leq w_{\max}$$

Cost is **linear in the number of assets** and effectively free in the number of scenarios, so the engine scales to dozens of holdings with several overlays at once. It is solved with the HiGHS LP solver, and a single set of common random numbers is reused across floors to give a smooth, comparable efficient frontier.

A note on comparability. The grid uses a fixed-threshold Expected Shortfall, $E[r_p | r_p < H]$, whereas the linear program uses α -CVaR. These coincide only when H equals the α -quantile; otherwise they are two valid but different tail measures. The scalable engine therefore **complements** the exact grid reference — it does not replace it.

6. Out-of-sample back-testing

Does the optimisation hold up on unseen data? The back-test builds optimal weights on a **construction window**, then holds those fixed weights (buy-and-hold) through a later **evaluation window**, marking any derivative **to market** at each step with a shrinking maturity. It compares what the model expected with what actually happened — realised versus expected return and volatility, and the loss-threshold outcome — and reports the realised **alpha, beta and R-squared** of each holding and of the portfolio against a chosen benchmark, with an optional CAPM (ex-ante) alpha. Both a no-derivative portfolio and a with-derivative portfolio are tracked side by side.

7. Validation

Every figure is checked rather than asserted:

- The Monte-Carlo optimum lies on the exact-grid mean-variance frontier to within a few basis points.
- Under the Gaussian copula the scenario Expected Shortfall matches the closed-form Normal value (for example, -28.4% versus -28.6%), confirming the generator and the tail estimate.
- The frontier is monotone, the ES floor binds at the optimum, and infeasible floors are flagged rather than silently relaxed.
- All thesis result tables reproduce within discretisation noise, end to end (R to Python).

Modelling horizon. Weights are optimised for a one-year hold (μ and σ are annualised), which lines up with an annual mandate, risk budget and yearly rebalancing; the back-test then bridges this single-period model horizon to a real, user-chosen holding window.

8. Implementation

The optimiser is written in Python (NumPy, SciPy, the HiGHS linear-program solver and copula sampling) behind a Streamlit interface with Plotly and Altair charts. An AI-assisted layer turns each result into a plain-language narrative and powers an interactive glossary of risk measures and portfolio theory. The numerical pipeline is open and reproducible: copula scenarios → Black–Scholes pricing → CVaR linear program → efficient frontier → interactive app and PDF report.

References

- Das, S. R. & Statman, M. (2009). Options and structured products in behavioral portfolios. Working paper / Journal of Economic Dynamics & Control.
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Research and educational project. Not investment advice. The interactive application reproduces all figures referenced above and exports per-run PDF reports.